Python: exercises on dictionaries, tuples, and sets

This notebook was generated in student mode.

Exercise 1: Merge Two Dictionaries

Exercise Description

Create a function `merge_dicts(dict1, dict2)` that combines two dictionaries. If a key exists in both, sum their values.

Your solution

```
def merge_dicts(dict1, dict2):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
dict1 = {'a': 1, 'b': 2}
dict2 = {'b': 3, 'c': 4}
result = merge_dicts(dict1, dict2)
print(result)  # Expected: {'a': 1, 'b': 5, 'c': 4}
```

Test your solution

```
# Test case 2
dict1 = {'x': 10}
dict2 = {'y': 20, 'x': 5}
result = merge_dicts(dict1, dict2)
print(result)  # Expected: {'x': 15, 'y': 20}
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 2: Reverse Lookup

Exercise Description

Write a function `reverse_lookup(d, value)` that returns all keys from dictionary `d` that map to the given `value`.

Your solution

```
def reverse_lookup(d, value):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
d = {'a': 1, 'b': 2, 'c': 1}
result = reverse_lookup(d, 1)
print(result)  # Expected: ['a', 'c']
```

Test your solution

```
# Test case 2
d = {'x': 'hello', 'y': 'world', 'z': 'hello'}
```

```python
result = reverse_lookup(d, 'hello')
print(result)  # Expected: ['x', 'z']
```

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 3: Count Tuple Elements

Exercise Description

Write a function `count_elements(tup)` that returns a dictionary with counts of each element in the tuple `tup`.

Your solution

```python
def count_elements(tup):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
tup = (1, 2, 2, 3, 1)
result = count_elements(tup)
print(result)  # Expected: {1: 2, 2: 2, 3: 1}
```

Test your solution

```
# Test case 2
tup = ('a', 'b', 'a', 'c', 'b', 'a')
result = count_elements(tup)
print(result)  # Expected: {'a': 3, 'b': 2, 'c': 1}
```

Debug your solution

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```

Exercise 4: Set Intersection

Exercise Description

Implement a function `intersect(s1, s2)` that returns the intersection of two sets `s1` and `s2`.

Your solution

```
def intersect(s1, s2):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
s1 = {1, 2, 3}
s2 = {2, 3, 4}
result = intersect(s1, s2)
print(result)  # Expected: {2, 3}
```

Test your solution

```
# Test case 2
s1 = {'a', 'b', 'c'}
s2 = {'b', 'c', 'd'}
result = intersect(s1, s2)
print(result)  # Expected: {'b', 'c'}
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 5: Remove Duplicates

Exercise Description

Write a function `remove_duplicates(lst)` that returns a list without any duplicates using a set.

Your solution

```
def remove_duplicates(lst):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
lst = [1, 2, 2, 3, 4, 4, 5]
result = remove_duplicates(lst)
print(result)  # Expected: [1, 2, 3, 4, 5]
```

Test your solution

```
# Test case 2
lst = ['a', 'b', 'a', 'c', 'b']
result = remove_duplicates(lst)
print(result)  # Expected: ['a', 'b', 'c'] (order may vary)
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 6: Tuple to Dictionary

Exercise Description

Create a function `tuple_to_dict(tup)` that turns a tuple of `(key, value)` pairs into a dictionary.

Your solution

```python
def tuple_to_dict(tup):
    # Write your solution here
    pass
```

Test your solution

```python
# Test case 1
tup = (('a', 1), ('b', 2))
result = tuple_to_dict(tup)
print(result)  # Expected: {'a': 1, 'b': 2}
```

Test your solution

```python
# Test case 2
tup = (('x', 10), ('y', 20), ('z', 30))
result = tuple_to_dict(tup)
print(result)  # Expected: {'x': 10, 'y': 20, 'z': 30}
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 7: Value Switch in Dictionary

Exercise Description

Write a function `switch_values(d)` that switches the keys and values in a dictionary. Assume all values are unique.

Your solution

```
def switch_values(d):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
d = {'a': 1, 'b': 2}
result = switch_values(d)
print(result)  # Expected: {1: 'a', 2: 'b'}
```

Test your solution

```
# Test case 2
d = {'name': 'John', 'age': 25}
result = switch_values(d)
print(result)  # Expected: {'John': 'name', 25: 'age'}
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 8: Dictionary of Squares

Exercise Description

Create a function `squares(n)` that returns a dictionary of numbers from 1 to `n` where the keys are numbers and the values are their squares.

Your solution

```
def squares(n):
    # Write your solution here
    pass
```

Test your solution

```python
# Test case 1
result = squares(5)
print(result)  # Expected: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Test your solution

```python
# Test case 2
result = squares(3)
print(result)  # Expected: {1: 1, 2: 4, 3: 9}
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 9: Set Difference

Exercise Description

Write a function `difference(s1, s2)` that returns a set of elements that are only in the first set `s1` but not in the second set `s2`.

Your solution

```
def difference(s1, s2):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
s1 = {1, 2, 3, 4}
s2 = {3, 4, 5, 6}
result = difference(s1, s2)
print(result)  # Expected: {1, 2}
```

Test your solution

```
# Test case 2
s1 = {'a', 'b', 'c'}
s2 = {'b', 'd'}
result = difference(s1, s2)
print(result)  # Expected: {'a', 'c'}
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 10: Count Vowels

Exercise Description

Create a function `count_vowels(s)` that counts and returns a dictionary with vowels as keys and their counts in the string `s` as values.

Your solution

```
def count_vowels(s):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
result = count_vowels('hello world')
print(result)  # Expected: {'e': 1, 'o': 2}
```

Test your solution

```
# Test case 2
result = count_vowels('programming')
print(result)  # Expected: {'o': 1, 'a': 1, 'i': 1}
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 11: Find Max Value

Exercise Description

Write a function `find_max(d)` that returns the key with the maximum value in the dictionary `d`.

Your solution

```
def find_max(d):
    # Write your solution here
```

```
    pass
```

Test your solution

```
# Test case 1
d = {'a': 5, 'b': 12, 'c': 3}
result = find_max(d)
print(result)  # Expected: 'b'
```

Test your solution

```
# Test case 2
d = {'x': 100, 'y': 50, 'z': 200}
result = find_max(d)
print(result)  # Expected: 'z'
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 12: Unique Characters

Exercise Description

Create a function `unique_chars(s)` that returns a set of unique characters in the string `s`.

Your solution

```
def unique_chars(s):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
result = unique_chars('hello')
print(result)  # Expected: {'e', 'h', 'l', 'o'}
```

Test your solution

```
# Test case 2
result = unique_chars('programming')
print(result)  # Expected: {'p', 'r', 'o', 'g', 'a', 'm', 'i', 'n'}
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the

Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 13: Key Multiplication

Exercise Description

Write a function `multiply_keys(d, factor)` that returns a new dictionary where each key is multiplied by a `factor`.

Your solution

```
def multiply_keys(d, factor):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1
d = {1: 100, 2: 200}
result = multiply_keys(d, 2)
print(result)  # Expected: {2: 100, 4: 200}
```

Test your solution

```
# Test case 2
d = {3: 'a', 5: 'b'}
result = multiply_keys(d, 10)
print(result)  # Expected: {30: 'a', 50: 'b'}
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 14: Set Complement

Exercise Description

Implement a function `complement(full_set, sub_set)` that returns the set of elements in `full_set` that are not in `sub_set`.

Your solution

```python
# Write your solution here
```

Test your solution

```python
# Test case 1
full_set = {1, 2, 3, 4, 5}
sub_set = {2, 4}
result = complement(full_set, sub_set)
print(result)  # Expected: {1, 3, 5}
```

Test your solution

```python
# Test case 2
full_set = {'a', 'b', 'c', 'd'}
sub_set = {'b', 'd'}
result = complement(full_set, sub_set)
print(result)  # Expected: {'a', 'c'}
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise 15: Count Keys Starting With

Exercise Description

Write a function `count_keys_starting_with(d, char)` that returns the count of keys in dictionary `d` that start with `char`.

Your solution

```
# Write your solution here
```

Test your solution

```
# Test case 1
d = {'apple': 1, 'banana': 2, 'apricot': 3, 'cherry': 4}
result = count_keys_starting_with(d, 'a')
print(result)  # Expected: 2
```

Test your solution

```
# Test case 2
d = {'hello': 1, 'world': 2, 'python': 3}
result = count_keys_starting_with(d, 'h')
print(result)  # Expected: 1
```


Debug your solution